Quinn Asena

Data Scientist & Ecologist

Cary Institute of Ecosystem Studies

Forest Futures Lab, Millbrook, NY

CONTACT

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LAB AFFILIATIONS

Forest Futures Lab
Cary Institute

Williams Paleoecology
UW-Madison

Perry Lab
U. Auckland

Neotoma Paleoecology DB
Leadership Council

METHODS & DATA

State-space time-series models
Age-depth modelling (Bchron)
Pseudoproxy / virtual ecology
Machine learning for spatial data
Neotoma database & standards
FAIR/CARE data stewardship

SKILLS & TOOLS

R · Python · C++ · Bash · SQL · Apache Arrow · Git/GitHub · Quarto · Shiny · HPC (Slurm/PBS)

INTERESTS

Classical piano · Rock climbing · Art

PROFILE

I am an **ecologist and data scientist** focused on how ecosystems change over time and space, and on building the data infrastructure needed to study them. My current work includes deep learning methods for **wildfire risk mapping**, and process-based **landscape simulations** to project forest biodiversity change. My PhD and postdoctoral research focused on palaeoecology and statistical method development to analyse irregular time series data. I currently serve as early career representative on the **Neotoma Paleoecology Database Leadership Council**.

PROFESSIONAL EXPERIENCE

Data Scientist — *Millbrook, NY*
Cary Institute of Ecosystem Studies — *2025 – Present*

- Automated HPC workflows for process-based landscape simulations across boreal North America under multiple CMIP6 climate scenarios.
- Large-scale data management using SQL and Apache Arrow; integration of climate, soil, and elevation data from distributed archives (NASA DAACs, STAC).
- Applying deep learning models to project wildfire risk across boreal North America.

Postdoctoral Research Associate — *Madison, WI*
University of Wisconsin-Madison — *2022 – 2024*

- NSF-funded project developing state-space modelling methods to analyse palaeoecological records.
- Abrupt change in ecosystems project led by Jack Williams and Anthony Ives.

Engagement Specialist, *Centre for eResearch - University of Auckland*: Delivered seminars on data management, FAIR and CARE principles, and reproducible workflows; provided research-computing support for graduate students and research staff (virtual machines, data storage, analytical workflows). *2021 – 2022*

Data Analyst, *Auckland Council*: Analysed citizen science bird count data and produced a technical report. *2021*

Research Assistant, *University of Auckland*: Conducted a bibliometric analysis on the topic of climate justice. *2021 – 2022*

Research Assistant, *Stockholm Environment Institute*: Reconstruction of fire regimes in the North York Moors. *2016 – 2017*

EDUCATION

PhD, University of Auckland, *School of Environment*: Explored virtual ecological methods for generating pseudoproxy data to assess statistical inferences under data uncertainty. Supervised by George Perry and Janet Wilmshurst. *2017 – 2021*

MEnv. Environmental Science, *University of York*: First-class integrated master's degree in environmental science. Master's research involved environmental monitoring using sensor networks and robotics. *2012 – 2016*

Geocomputation & Machine Learning for Environmental Applications, *Basilicata University*: Certified course in machine learning for spatial analyses. *2025*

Certified Carpentries Instructor, *The Carpentries*: Certified to teach and host Carpentries workshops in data science and computational skills. *2023*

PUBLICATIONS

- Perrotti, A.G., Williams, J.W., **Asena, Q.**, Schlenker, N., Lam, J., Siedelmann, M., Russell, J.M., Ives, A.R. (In review). *Past novel ecosystems in Florida driven by fluctuating climate, fire, and megaherbivory.*
- Johnson, J.W., Ray-Cozzens, T., Shuman, B.N., Williams, J.W., Ives, A.R., **Asena, Q.**, Stefanova, V., Nelson, D.M. (In review). *Differential responses of American Beech and Eastern Hemlock to Holocene drought and warming: Insights from pollen $\delta^{13}C$ and multiproxy reconstructions.*
- **Asena, Q.**, Williams, J.W., Johnson, J.W., Shuman, B.N., Stefanova, V., Ives, A.R. (2026). Statistical analyses of ecological multinomial time series to identify environmental drivers and biotic interactions. *Methods in Ecology and Evolution.*
- **Asena, Q.**, Perry, G.L., Wilmshurst, J.M. (2026). Information loss in palaeoecological data from process and observer error. *Climate of the Past.*
- Lee, F., Bellvé, A.M., Alena, H., **Asena, Q.**, Perry, G.L. (2025). Using population viability analysis and fossil records to inform the conservation of pateke (*Anas chlorotis*). *New Zealand Journal of Ecology.*
- **Asena, Q.**, Perry, G.L., Wilmshurst, J.M. (2024). Is the past recoverable from the data? Pseudoproxy modelling of uncertainties in palaeoecological data. *The Holocene.*
- Parsons, M., **Asena, Q.**, Johnson, D., Nalau, J. (2024). A Bibliometric Review and Topic Analysis of Climate Justice Literature: Mapping Trends, Voices, and the Way Forward. *Climate Risk Management.* Code repository publicly available.
- Courtney-Mustaphi, C.J., Brahney, J., Aquino-López, M.A., Goring, S., Orton, K., Noronha, A., Czaplewski, J., **Asena, Q.**, Paton, S. (2019). Guidelines for Reporting and Archiving ^{210}Pb Sediment Chronologies to Improve Fidelity and Extend Data Lifecycle. *Quaternary Geochronology.*
- Heinemeyer, A., Burn, W.L., **Asena, Q.**, Jones, A., Ashby, M. (2019). Response to comment on Peatland carbon stocks and burn history. *GEO: Geography and Environment.*
- Heinemeyer, A., **Asena, Q.**, Burn, W.L., Jones, A. (2018). Peatland carbon stocks and burn history: Blanket bog peat core evidence highlights charcoal impacts on peat physical properties and long-term carbon storage. *GEO: Geography and Environment.*
- Simpkins, C.E., Lee, F., Powers, B.F., Anderson, S., **Asena, Q.**, Brock, J.M.R., Perry, G.L. (2018). Population viability analyses in New Zealand: a review. *New Zealand Journal of Ecology.*
- Hamilton, R., Brussel, T., **Asena, Q.**, Bruel, R., Marcisz, K., Słowiński, M., Morris, J. (2018). Assessing the links between resilience, disturbance, and functional traits in paleoecological datasets. *Past Global Changes Magazine, vol. 26(2), 87.*

SOFTWARE & OPEN SCIENCE

Palaeoecological data science: a data-driven curriculum: an open-source book for teaching palaeoecological data science.

multinomialTS: R package for state-space modelling of multinomially distributed ecological time-series data.

State-Space Methods for Palaeoecological Hypothesis Testing: Open workshop materials for state-space modelling of community and palaeoecological time series.

Authoring Collaborative Research Projects in Quarto: Resources for collaborative research workflows using Git, GitHub, and Quarto (ResBaz 2022–2024).

fisheR: R package for calculating Fisher's Information on ecological time-series data.

Population Growth Explorer: Interactive app exploring population growth equations for undergraduate teaching.

SERVICE & LEADERSHIP

Early Career Representative, *Neotoma Leadership Council*: Contributing to database governance, ingestion standards, and community user support for a global palaeoecological data resource. *2024 – Present*

Committee member, *Computational Resources Committee*: Contributing to data infrastructure design and decisions, including security, data management, and FAIR/CARE practices. *2025 – Present*

Committee member, *Postdoc, Research Associate, and Data Analyst/Data Scientist (PRADAS) committee*: Sharing information, training, and skills across the Cary PRADAS community. *2026 – Present*

SELECTED TALKS, WORKSHOPS & CONFERENCES

Methods in Ecology and Evolution, *Virtual*; Talk: multinomialTS: a state-space approach for modelling community time series. Materials: Talk and workshop resources. *2026*

ESA 2026, *Salt Lake City, UT*; **Speaker**: Assessing climate-driven compositional shifts in boreal forests using iLand. *2026*

Neotoma Outreach, *Virtual*; Talk: A virtual ecological approach to modelling uncertainties in palaeoecology. Workshop: Hands-on workshop for age-depth modelling using Bchron. *2026*

ESA 2025, *Baltimore, MD*; **Speaker**: Modelling palaeoecological datasets using a state-space approach: two case studies. *2025*

AGU, *New Orleans, LA*; Poster: Assessing climate-driven compositional shifts in boreal forests using iLand. *2025*

ESA 2024, *Long Beach, CA*; **Speaker**: Hypothesis testing of the drivers of long-term ecological change in Michigan. **Workshop host**: State-space methods for testing hypotheses about environmental drivers and biotic interactions. *2024*

INQUA 2023, *Rome, Italy*; **Lead Convenor**: Data science and palaeoecology: current intersections and advances. *2023*

ESA 2023, *Portland, OR*; **Speaker**: Modelling palaeoecological community data: a state-space approach. *2023*

ResBaz, *Auckland, NZ; Madison, WI*; Organiser: ResBaz Aotearoa 2021. Workshop lead: Quarto for collaboration at ResBaz NZ 2022–2024 and in-person Madison, WI, 2023. *2021 – 2025*

AGU, *Chicago, IL*; Speaker: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudoproxies. *2022*

AMQUA, *Madison, WI*; Poster: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudoproxies. *2022*

BES Palaeo in R, *Virtual*; Speaker: Virtual ecological methods for assessing uncertainty in palaeoecological data using pseudoproxies. *2022*

INQUA 2019, *Dublin, Ireland*; Speaker: Pseudoproxy modelling for assessing statistical inferences. *2019*

International Swiss Climate Summer School, *University of Bern, Bern, Switzerland*; Poster presentation and participant. Accredited course (3 ECTS credits). *2018*

EcoRe3: Resistance Recovery and Resilience in Long-term Ecological Systems, *PAGES - University of Utah, Utah, USA*; Speaker: modelling resilience in ecosystems. *2018*

TEACHING

PhD Student Mentor, *SUNY*: One-on-one mentoring in statistics for palaeoecology. *2025*

Graduate Teaching Assistant & Assistant Coordinator, *University of Auckland*: Assistant coordinator for Discovering Environmental Modelling. Guest lecturer on population growth. Developed a Shiny application as a lecture aid. *2018 – 2020*

PEER REVIEW

Reviewer for *Ecology Letters*, *Journal of Open Source Software*, *Nature Communications*, *Environmental Research and Technology*, *Quaternary Geochronology*, and *Methods in Ecology and Evolution*.